



Functions (1797) and *Lessons on the Calculus of Functions* (1801). These works had a great influence on the evolution of calculus in the first third of the nineteenth century.

In appearance, Lagrange was of medium height and slightly formed, with pale blue eyes and a colorless complexion. His character, in the words of mathematician W. W. Rouse Ball, was "nervous and timid; he detested controversy, and to avoid it willingly allowed others to take the credit for what he had himself done."

7.2 TRIGONOMETRIC INTEGRALS

In this section, we examine integrals of functions that are products of powers of the trigonometric functions. In particular, we consider integrals of the form

$$\int \sin^m x \cos^n x \, dx,$$

where m and n are integers. In Example 7 of Section 7.1, we obtained a reduction formula for $\int \sin^n x \, dx$. Integrals of this type may also be found without using integration by parts. If n is an odd positive integer, we begin by writing

$$\int \sin^n x \, dx = \int \sin^{n-1} x \sin x \, dx.$$

Since the integer $n - 1$ is even, we may then use the trigonometric identity $\sin^2 x = 1 - \cos^2 x$ to obtain a form that is easy to integrate, as illustrated in the next example.

EXAMPLE ■ 1 Evaluate $\int \sin^5 x \, dx$.

SOLUTION As in the preceding discussion, we write

$$\begin{aligned} \int \sin^5 x \, dx &= \int \sin^4 x \sin x \, dx \\ &= \int (\sin^2 x)^2 \sin x \, dx \\ &= \int (1 - \cos^2 x)^2 \sin x \, dx \\ &= \int (1 - 2\cos^2 x + \cos^4 x) \sin x \, dx. \end{aligned}$$

If we substitute

$$u = \cos x, \quad du = -\sin x \, dx,$$

we obtain

$$\begin{aligned} \int \sin^5 x \, dx &= - \int (1 - 2\cos^2 x + \cos^4 x)(-\sin x) \, dx \\ &= - \int (1 - 2u^2 + u^4) \, du \\ &= -u + \frac{2}{3}u^3 - \frac{1}{5}u^5 + C \\ &= -\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C. \end{aligned}$$

Similarly, for odd powers of $\cos x$, we write

$$\int \cos^n x \, dx = \int \cos^{n-1} x \cos x \, dx$$

and use the fact that $\cos^2 x = 1 - \sin^2 x$ to obtain an integrable form.

If the integrand is $\sin^n x$ or $\cos^n x$ and n is even, then the half-angle formula

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \text{or} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

may be used to simplify the integrand.

EXAMPLE ■ 2 Evaluate $\int \cos^2 x \, dx$.

SOLUTION Using a half-angle formula, we have

$$\begin{aligned} \int \cos^2 x \, dx &= \frac{1}{2} \int (1 + \cos 2x) \, dx \\ &= \frac{1}{2}x + \frac{1}{4}\sin 2x + C. \end{aligned}$$

EXAMPLE ■ 3 Evaluate $\int \sin^4 x \, dx$.

SOLUTION

$$\begin{aligned} \int \sin^4 x \, dx &= \int (\sin^2 x)^2 \, dx \\ &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 \, dx \\ &= \frac{1}{4} \int (1 - 2\cos 2x + \cos^2 2x) \, dx \end{aligned}$$

We apply a half-angle formula again and write

$$\cos^2 2x = \frac{1}{2}(1 + \cos 4x) = \frac{1}{2} + \frac{1}{2}\cos 4x.$$

Substituting in the last integral and simplifying gives us

$$\begin{aligned}\int \sin^4 x \, dx &= \frac{1}{4} \int \left(\frac{3}{2} - 2 \cos 2x + \frac{1}{2} \cos 4x \right) dx \\ &= \frac{3}{8}x - \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + C.\end{aligned}$$

Integrals involving only products of $\sin x$ and $\cos x$ may be evaluated using the following guidelines.

Guidelines for Evaluating

$$\int \sin^m x \cos^n x \, dx \quad 7.2$$

- 1 If m is an odd integer: Write the integral as

$$\int \sin^m x \cos^n x \, dx = \int \sin^{m-1} x \cos^n x \sin x \, dx$$

and express $\sin^{m-1} x$ in terms of $\cos x$ by using the trigonometric identity $\sin^2 x = 1 - \cos^2 x$. Make the substitution

$$u = \cos x, \quad du = -\sin x \, dx$$

and evaluate the resulting integral.

- 2 If n is an odd integer: Write the integral as

$$\int \sin^m x \cos^n x \, dx = \int \sin^m x \cos^{n-1} x \cos x \, dx$$

and express $\cos^{n-1} x$ in terms of $\sin x$ by using the trigonometric identity $\cos^2 x = 1 - \sin^2 x$. Make the substitution

$$u = \sin x, \quad du = \cos x \, dx$$

and evaluate the resulting integral.

- 3 If m and n are even: Use half-angle formulas for $\sin^2 x$ and $\cos^2 x$ to reduce the exponents by one-half.

EXAMPLE 4 Evaluate $\int \cos^3 x \sin^4 x \, dx$.

SOLUTION By guideline (2) of (7.2),

$$\begin{aligned}\int \cos^3 x \sin^4 x \, dx &= \int \cos^2 x \sin^4 x \cos x \, dx \\ &= \int (1 - \sin^2 x) \sin^4 x \cos x \, dx.\end{aligned}$$

If we let $u = \sin x$, then $du = \cos x \, dx$, and the integral may be written

$$\begin{aligned}\int \cos^3 x \sin^4 x \, dx &= \int (1 - u^2)u^4 \, du = \int (u^4 - u^6) \, du \\ &= \frac{1}{5}u^5 - \frac{1}{7}u^7 + C \\ &= \frac{1}{5}\sin^5 x - \frac{1}{7}\sin^7 x + C.\end{aligned}$$

The following guidelines are analogous to those in (7.2) for integrands of the form $\tan^m x \sec^n x$.

Guidelines for Evaluating

$\int \tan^m x \sec^n x dx$ **7.3**

- 1 If m is an odd integer:** Write the integral as

$$\int \tan^m x \sec^n x dx = \int \tan^{m-1} x \sec^{n-1} x \sec x \tan x dx$$

and express $\tan^{m-1} x$ in terms of $\sec x$ by using the trigonometric identity $\tan^2 x = \sec^2 x - 1$. Make the substitution

$$u = \sec x, \quad du = \sec x \tan x dx$$

and evaluate the resulting integral.

- 2 If n is an even integer:** Write the integral as

$$\int \tan^m x \sec^n x dx = \int \tan^m x \sec^{n-2} x \sec^2 x dx$$

and express $\sec^{n-2} x$ in terms of $\tan x$ by using the trigonometric identity $\sec^2 x = 1 + \tan^2 x$. Make the substitution

$$u = \tan x, \quad du = \sec^2 x dx$$

and evaluate the resulting integral.

- 3 If m is even and n is odd:** There is no standard method of evaluation. Possibly use integration by parts.

EXAMPLE 5 Evaluate $\int \tan^3 x \sec^5 x dx$.

SOLUTION By guideline (1) of (7.3),

$$\begin{aligned} \int \tan^3 x \sec^5 x dx &= \int \tan^2 x \sec^4 x (\sec x \tan x) dx \\ &= \int (\sec^2 x - 1) \sec^4 x (\sec x \tan x) dx. \end{aligned}$$

Substituting $u = \sec x$ and $du = \sec x \tan x dx$, we obtain

$$\begin{aligned} \int \tan^3 x \sec^5 x dx &= \int (u^2 - 1) u^4 du \\ &= \int (u^6 - u^4) du \\ &= \frac{1}{7} u^7 - \frac{1}{5} u^5 + C \\ &= \frac{1}{7} \sec^7 x - \frac{1}{5} \sec^5 x + C. \end{aligned}$$

Exercises 7.3

EXAMPLE 4

Evaluate $\int \tan^3 x \sec^2 x \, dx$.

SOLUTION

By problem 12 of 7.2,

$$\begin{aligned}\int \tan^3 x \sec^2 x \, dx &= \int \tan^2 x \sec^2 x \sec^2 x \, dx \\ &= \int \tan^2 x (\tan^2 x + 1) \sec^2 x \, dx.\end{aligned}$$

If we let $u = \tan x$, then $du = \sec^2 x \, dx$, and

$$\begin{aligned}\int \tan^3 x \sec^2 x \, dx &= \int u^2(u^2 + 1) \, du \\ &= \int (u^4 + u^2) \, du \\ &= \frac{1}{5}u^5 + \frac{1}{3}u^3 + C \\ &= \frac{1}{5}\tan^5 x + \frac{1}{3}\tan^3 x + C.\end{aligned}$$

Integrals of the form $\int \sec^m x \tan^n x \, dx$ may be evaluated in similar fashion.

Example: If the integral has only odd powers of $\tan x$, we can remove one factor of $\tan x$ and use the identity $\tan^2 x = \sec^2 x - 1$ to reduce the power of $\tan x$ by 2. For example, to evaluate

EXAMPLE 5

Evaluate $\int \sec^3 2x \tan 2x \, dx$.

Using the reduction formula for $\sec^m x$, we obtain

$$\begin{aligned}\int \sec^3 2x \tan 2x \, dx &= \int \frac{1}{2} \sec^2 2x \cdot \tan 2x \, dx \\ &= \frac{1}{2} \sec 2x + \frac{1}{2} \tan 2x + C.\end{aligned}$$

EXERCISES 7.3

Exer. 1–10: Evaluate the integral.

1. $\int \tan^2 x \, dx$

2. $\int \sec^2 x \tan^2 x \, dx$

3. $\int \tan^4 x \, dx$

4. $\int \tan^6 x \, dx$

5. $\int \sin^2 x \cos^2 x \, dx$

6. $\int \sin^4 x \, dx$

7. $\int \sin^3 x \cos^2 x \, dx$

8. $\int \cos^3 x \sin^2 x \, dx$

- 9 $\int \tan^3 x \sec^4 x \, dx$ 10 $\int \sec^6 x \, dx$
 11 $\int \tan^3 x \sec^3 x \, dx$ 12 $\int \tan^5 x \sec x \, dx$
 13 $\int \tan^6 x \, dx$ 14 $\int \cot^4 x \, dx$
 15 $\int \sqrt{\sin x} \cos^3 x \, dx$ 16 $\int \frac{\cos^3 x}{\sqrt{\sin x}} \, dx$
 17 $\int (\tan x + \cot x)^2 \, dx$ 18 $\int \cot^3 x \csc^3 x \, dx$
 19 $\int_0^{\pi/4} \sin^3 x \, dx$ 20 $\int_0^1 \tan^2(\frac{1}{3}\pi x) \, dx$
 21 $\int \sin 5x \sin 3x \, dx$ 22 $\int_0^{\pi/4} \cos x \cos 5x \, dx$
 23 $\int_0^{\pi/2} \sin 3x \cos 2x \, dx$ 24 $\int \sin 4x \cos 3x \, dx$
 25 $\int \csc^4 x \cot^4 x \, dx$ 26 $\int (1 + \sqrt{\cos x})^2 \sin x \, dx$
 27 $\int \frac{\cos x}{2 - \sin x} \, dx$ 28 $\int \frac{\tan^2 x - 1}{\sec^2 x} \, dx$
 29 $\int \frac{\sec^2 x}{(1 + \tan x)^2} \, dx$ 30 $\int \frac{\sec x}{\cot^5 x} \, dx$
- 31 The region bounded by the x -axis and the graph of $y = \cos^2 x$ from $x = 0$ to $x = 2\pi$ is revolved about the x -axis. Find the volume of the resulting solid.
- 32 The region between the graphs of $y = \tan^2 x$ and $y = 0$ from $x = 0$ to $x = \pi/4$ is revolved about the x -axis. Find the volume of the resulting solid.

33 The velocity (at time t) of a point moving on a coordinate line is $\cos^2 \pi t$ ft/sec. How far does the point travel in 5 sec?

34 The acceleration (at time t) of a point moving on a coordinate line is $\sin^2 t \cos t$ ft/sec². At $t = 0$, the point is at the origin and its velocity is 10 ft/sec. Find its position at time t .

35 (a) Prove that if m and n are positive integers,

$$\int \sin mx \sin nx \, dx = \begin{cases} \frac{\sin(m-n)x}{2(m-n)} - \frac{\sin(m+n)x}{2(m+n)} + C & \text{if } m \neq n \\ \frac{x}{2} - \frac{\sin 2mx}{4m} + C & \text{if } m = n \end{cases}$$

(b) Obtain formulas similar to that in part (a) for

$$\int \sin mx \cos nx \, dx$$

and $\int \cos mx \cos nx \, dx.$

36 (a) Use part (a) of Exercise 35 to prove that

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

(b) Find

(i) $\int_{-\pi}^{\pi} \sin mx \cos nx \, dx$

(ii) $\int_{-\pi}^{\pi} \cos mx \cos nx \, dx$

7.3

TRIGONOMETRIC SUBSTITUTIONS

In Example 4 on p. 42, we saw how to change an expression of the form $\sqrt{a^2 - x^2}$, with $a > 0$, into a trigonometric expression without radicals, by using the *trigonometric substitution* $x = a \sin \theta$. We can use a similar procedure for $\sqrt{a^2 + x^2}$ and $\sqrt{x^2 - a^2}$. This technique is useful for eliminating radicals from certain types of integrands. The substitutions are listed in the following table.

Trigonometric Substitutions 7.4

Expression in integrand	Trigonometric substitution
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$